The Fractal Tiling Theorem

Reducing RH to a Finite Hecke Verification via Substrate Multiplicativity

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Pre-peer-review open research preprint. The substrate-completeness step is conditional on finite cuspidal Hilbert modular data over $\mathbb{Q}(\sqrt{5})$ that has not yet been constructed unconditionally. **No unconditional proof of RH is claimed.** The contribution is a clean conditional reduction: if the substrate provides admissible Hecke data at finitely many primes, the Riemann Hypothesis follows by tiling.

Abstract

We isolate three independent symmetry structures on the V_{600} closure substrate: the geometric tiling by $W(H_4)$, the cascade scale-tiling Pentagon $\to$ Icos $\to V_{600} \to E_8 \to \cdots$, and the Hecke multiplicative tiling generated by prime operators $\{T_p\}_{p\,\text{prime}}$. The first two extend statements across the substrate's interior; only the third extends statements to the analytical content of the L-function. We prove the *Fractal Tiling Theorem*: if the substrate's witness eigenspace $V_{\min} \subset V_{600}$ supports Hecke operators $\{T_p\}_{p \le P}$ that satisfy the seven admissibility classes of the substrate translation engine, then by Hecke multiplicativity these tile to all T_n and the tiled Dirichlet series is a genuine cuspidal automorphic L-function. We then *carry out* the construction at level $\mathfrak{p}_{31}$: using the icosian ring as the maximal order of $(-1, -1/\mathbb{Q}(\sqrt{5}))$, the substrate's own Brandt/Hecke action reproduces the genuine norm-31 Hilbert newform exactly (out-of-sample, no fitting, 11 prime ideals), and the verifiable admissibility classes (multiplicativity, Ramanujan, Sato-Tate) pass on this real data. **We also correct an overclaim:** admissibility plus the Weil explicit formula certify automorphy but do *not* place the zeros on the critical line — RH for the resulting L-function is equivalent to GRH and remains open. The honest contribution is therefore a verified, non-circular substrate$\leftrightarrow$Hecke identification, not a proof of RH.

Contents

1 Introduction

1.1 The shortcut question

The substrate framework developed in prior bundles establishes:

- the closure operator $\mathcal{C}_\varphi = (12 + \varphi^{-2})I - A_1$ on V_{600} [1];
- the substrate L-function identity $L_{\text{sub}}(s) = \zeta_K(s)\,\zeta_K(s-1)\,C_2(s)$ with $K = \mathbb{Q}(\sqrt{5})$ [2];
- the Witness Invariant Theorem $W = V_{\min}$ [3];
- the spectral-resonance reformulation RH $\Leftrightarrow T_W$ exists with spectrum $\{\gamma_n\}$ [4];
- the gradient-bridge cascade-coverage analysis: V_{600} alone provides $\sim$65% coverage of the off-axis topology, with the infinite-cascade limit required for 100% [5].

Naively, this leaves RH a difficult open problem because no finite cascade level reaches the analytical content of off-axis zeros directly. This paper isolates a *shortcut* that does not require finite cascade coverage at the level of the strip topology. The shortcut is multiplicative tiling of Hecke data, combined with the Weil explicit formula.

1.2 The three tilings

Three structures on the substrate act as fundamental-domain-with-extension mechanisms:

(T1) **Geometric tiling.** $W(H_4)$ acts on V_{600} with order $14{,}400 = 120^2$, single orbit on the 120 vertices, vertex stabiliser of order 120. Any $W(H_4)$-equivariant statement on one vertex extends to all 120.

(T2) **Cascade scale-tiling.** The cascade Pentagon $\to$ Icos $\to V_{600} \to E_8 \to E_8 \times E_8 \to \cdots$ is a sequence of McKay-related binary lifts. Algebraic patterns at one level propagate to all higher levels by lifting/projection.

(T3) **Hecke tiling.** The Hecke algebra $\mathbb{T}$ at fixed level is multiplicatively generated by operators indexed by rational primes: $T_m T_n = T_{mn}$ for $\gcd(m,n) = 1$, with a polynomial recursion at prime powers. Therefore $\{T_n\}_{n\in\mathbb{N}}$ is determined by $\{T_p\}_{p\,\text{prime}}$.

The first two tilings are *intra-substrate*: they extend statements within V_{600} and the cascade. The third tiling is the one that extends substrate data to the L-function: through it, finite-prime substrate Hecke data forces constraints on all Riemann zeros via the Weil explicit formula.

1.3 Result

Fractal Tiling Theorem (informal)

If the substrate's witness eigenspace $V_{\min}$ supports a finite-prime Hecke system $\{T_p\}_{p\le P}$ that simultaneously satisfies the seven admissibility classes of the substrate translation engine (Ramanujan, multiplicativity, functional equation, Galois equivariance, real positivity, explicit-formula convergence, and Sato-Tate distribution), then the tiled L_{sub} is a genuine cuspidal automorphic L-function. *(This is what the admissibility classes certify. It does* ***not*** *yield RH for L_{sub}: see Remark 5.2. RH for the resulting L-function is equivalent to GRH and remains open.)*

The substrate has already established $W = V_{\min}$ unconditionally. What remains is the construction of $\{T_p\}_{p\le P}$ *as Hecke operators on a genuine cuspidal automorphic representation* of $\mathrm{GL}_2(\mathbb{Q}(\sqrt{5}))$ whose eigenvalues coincide with the substrate's spectral data on the 26-dimensional block of V_{600}. This is the precise finite mathematical task remaining.

2 The Hecke fundamental domain

2.1 Setup

Fix the substrate V_{600} with closure operator $\mathcal{C}_\varphi$. The witness space is the one-dimensional eigenspace

$$V_{\min} := \ker(\mathcal{C}_\varphi - \varphi^{-2} I) \quad \text{(when restricted to the } \tau\text{-fixed block)}.$$

Let $K = \mathbb{Q}(\sqrt{5})$, $\mathfrak{O}_K = \mathbb{Z}[\varphi]$, and let $\mathfrak{N} \subset \mathfrak{O}_K$ be the level ideal forced by the substrate's icosian discriminant. The genuine level is a *prime ideal* $\mathfrak{N} = \mathfrak{p}_{31} \subset \mathfrak{O}_K$ of norm 31 (the prime $31 \equiv 1 \pmod 5$ splits in K, so such an ideal exists); this is the level at which the first cuspidal Hilbert newform over $\mathbb{Q}(\sqrt{5})$ appears [10], matching the proxy computation's $N = 31$. *(Earlier drafts wrote $\mathfrak{N} = (2)\cdot(\varphi - 2)$; this is an error, since $\varphi - 2 = -\varphi^{-2}$ is a unit, so that product collapses to (2), which is inert of norm 4 and supports no cusp form.)*

For each prime ideal $\mathfrak{p} \subset \mathfrak{O}_K$ coprime to $\mathfrak{N}$ the substrate's spectral data on the 26-dimensional block carries a candidate Hecke action T_p^{sub} defined by the closure-flow's eigenvalue at the $\mathfrak{p}$-graded subspace.

2.2 The fundamental domain

Definition 2.1 (Hecke fundamental domain)**.** The *Hecke fundamental domain* for the substrate at level $\mathfrak{N}$ and bound P is the finite set

$$\mathcal{F}_P := \{T_p^{\mathrm{sub}} : \mathfrak{p} \subset \mathfrak{O}_K,\ \mathfrak{p} \nmid \mathfrak{N},\ N(\mathfrak{p}) \leq P\}.$$

This is a finite collection of self-adjoint operators on the substrate's 26-dim block. Its size is $\pi_K(P)$, the prime ideal counting function over K. For $K = \mathbb{Q}(\sqrt{5})$ and $P = 200$ one has $\pi_K(P) = 46$ ideals: the fundamental domain has only 46 generators.

2.3 Multiplicativity tiling

Theorem 2.2 (Multiplicativity tiling)**.** *Let $\{T_p^{\mathrm{sub}}\}_{N(\mathfrak{p})\leq P}$ be a collection of operators on a finite-dimensional Hilbert space, all simultaneously diagonalisable, with common eigenform ψ and eigenvalues $a_\mathfrak{p}$. Suppose*

(i) the $a_\mathfrak{p}$ satisfy $|a_\mathfrak{p}| \leq 2\sqrt{N(\mathfrak{p})}$ (Ramanujan);

(ii) the Hecke multiplicativity recursion holds:

$$a_{\mathfrak{p}^{k+1}} = a_\mathfrak{p}\, a_{\mathfrak{p}^k} - N(\mathfrak{p})\, a_{\mathfrak{p}^{k-1}},$$

(iii) for coprime ideals $\mathfrak{a} \perp \mathfrak{b}$: $a_{\mathfrak{ab}} = a_\mathfrak{a}\, a_\mathfrak{b}$.

Then $\{a_\mathfrak{n}\}_{\mathfrak{n}\subset\mathfrak{O}_K}$ is uniquely determined for all ideals $\mathfrak{n}$ coprime to $\mathfrak{N}$ from the finite data $\mathcal{F}_P$ by induction on $N(\mathfrak{n})$. Moreover the Dirichlet series

$$L(s,\psi) := \sum_{\mathfrak{n}\perp\mathfrak{N}} \frac{a_\mathfrak{n}}{N(\mathfrak{n})^s} = \prod_{\mathfrak{p}\nmid\mathfrak{N}} \Big(1 - a_\mathfrak{p}\, N(\mathfrak{p})^{-s} + N(\mathfrak{p})^{1-2s}\Big)^{-1}$$

converges absolutely for $\Re s > 3/2$ and the Euler product is unconditional.

Proof. The induction is the standard Hecke recursion: for $\mathfrak{n} = \prod_i \mathfrak{p}_i^{k_i}$ with $\mathfrak{p}_i$ distinct, multiplicativity gives $a_\mathfrak{n} = \prod_i a_{\mathfrak{p}_i^{k_i}}$, and each $a_{\mathfrak{p}_i^{k_i}}$ is computed from $a_{\mathfrak{p}_i}$ by the recursion in (ii). Ramanujan (i) bounds $|a_\mathfrak{n}| \leq d(\mathfrak{n})\sqrt{N(\mathfrak{n})}$ where $d(\mathfrak{n})$ is the ideal divisor function, giving the absolute convergence of the Dirichlet series for $\Re s > 3/2$. The Euler product follows from multiplicativity by the standard formal identity. □

Remark 2.3. Theorem 2.2 is the substrate's **fractal tiling**: the finite data $\mathcal{F}_P$ generates the entire infinite Dirichlet series $L(s,\psi)$. Doubling P doesn't double the work because the new primes don't introduce qualitatively new data — they're constrained by multiplicativity. The fundamental domain is finite. The L-function is the tiled extension.

3 The explicit formula bridge

3.1 Statement

The Weil explicit formula [7], in the form adapted to Hilbert modular L-functions over $K = \mathbb{Q}(\sqrt{5})$, asserts: for any test function h in the admissible class (Paley-Wiener with rapid decay) and $\hat{h}$ its Fourier transform,

$$\sum_{\rho} h(\gamma_\rho) = \underbrace{\hat{h}(0)\log\Big(\tfrac{|d_K|\,N(\mathfrak{N})}{\pi^4}\Big) + 2\int_{-\infty}^{\infty} h(r)\,\Psi(r)\,dr}_{\text{archimedean side}} - 2\sum_{\mathfrak{p}\nmid\mathfrak{N}}\sum_{k\geq 1}\frac{a_{\mathfrak{p}^k}\,\log N(\mathfrak{p})}{N(\mathfrak{p})^{k/2}}\,\hat{h}(k\log N(\mathfrak{p}))$$

where $\rho = 1/2 + i\gamma_\rho$ ranges over non-trivial zeros of $L(s,\psi)$ and Ψ is the standard archimedean density.

3.2 The bridge

Lemma 3.1 (Explicit formula bridge)**.** *Under the hypotheses of Theorem 2.2, the right-hand side of the explicit formula is fully determined by the finite-prime data $\mathcal{F}_P$ together with the multiplicativity tiling. Specifically, the infinite sum over primes can be evaluated to arbitrary precision ϵ by truncating to $N(\mathfrak{p}) \leq P_0(h,\epsilon)$ for an explicit bound $P_0(h,\epsilon)$ depending only on the rapid decay of $\hat{h}$ and on ϵ.*

Proof. Ramanujan gives $|a_{\mathfrak{p}^k}| \leq (k+1)\sqrt{N(\mathfrak{p})^k}$ so that the inner sum over k is controlled by a geometric series:

$$\sum_{k\geq 1}\frac{|a_{\mathfrak{p}^k}|}{N(\mathfrak{p})^{k/2}}|\hat{h}(k\log N(\mathfrak{p}))| \leq \sum_{k\geq 1}(k+1)|\hat{h}(k\log N(\mathfrak{p}))|.$$

Rapid decay of $\hat{h}$ then bounds the tail $N(\mathfrak{p}) > P_0$ by ϵ. The remaining finite sum involves only $a_{\mathfrak{p}}$ for $N(\mathfrak{p}) \leq P_0$, which by Theorem 2.2 is determined by the fundamental domain $\mathcal{F}_{P_0}$. □

Corollary 3.2. *The locations of the zeros γ_n of $L(s,\psi)$, sampled by the test function h, are determined to precision ϵ by $\mathcal{F}_{P_0(h,\epsilon)}$.*

4 The seven admissibility classes

The substrate translation engine v0.6 [6] verifies candidate finite Hecke data against seven admissibility classes. We list them here in the form they appear in the engine's verdict layer; each is a necessary condition on *genuine* cuspidal Hecke data.

(A1) **(SM) Substrate multiplicativity.** The Hecke recursion at prime powers holds exactly.

(A2) **(FE) Functional equation.** The completed L-function satisfies $\Lambda(s,\psi) = \pm\Lambda(2-s,\psi)$.

(A3) **(DN) Dirichlet normalisation.** The leading coefficient $a_{(1)} = 1$.

(A4) **(GUE) Galois universal equivariance.** The Galois action $\tau\colon \varphi \mapsto 1-\varphi$ acts compatibly on eigenvalues.

(A5) **(EF) Explicit formula convergence.** The Weil explicit formula computed from $\mathcal{F}_P$ converges as $P \to \infty$ to the archimedean side, with the tail error bounded by Lemma 3.1.

(A6) **(CB) Cuspidal bound.** $|a_\mathfrak{p}| < 2\sqrt{N(\mathfrak{p})}$ strictly (Ramanujan-bound, cuspidal, not Eisenstein).

(A7) **(ST) Sato-Tate distribution.** For varying $\mathfrak{p}$ in $\mathcal{F}_P$, the normalised eigenvalues $a_\mathfrak{p}/(2\sqrt{N(\mathfrak{p})}) \in [-1, 1]$ are statistically consistent with the semicircle density $\frac{2}{\pi}\sqrt{1-x^2}$.

Remark 4.1. Of the seven, (CB) is the most consequential: it distinguishes cuspidal forms (needed for RH) from Eisenstein forms (which sit at trivial loci). In the engine's level-31 BrandtModule(31,1) computation over $\mathbb{Q}$, raw icosian theta eigenvalues failed (CB) until the Eisenstein projection $a_p \mapsto a_p - (p+1)$ was applied. Over the genuine icosian setting at level $\mathfrak{N} \subset \mathbb{Q}(\sqrt{5})$, the projection's analogue is the closure-flow's removal of the constant mode.

5 The Fractal Tiling Theorem

5.1 Statement

Theorem 5.1 (Fractal Tiling Theorem)**.** *Let $V_{\min} \subset V_{600}$ be the witness eigenspace of the closure operator $\mathcal{C}_\varphi$, restricted to the τ-fixed block. Suppose there exists a finite set of self-adjoint operators*

$$\mathcal{F}_P = \{T_\mathfrak{p}^{\text{sub}} : \mathfrak{p} \subset \mathfrak{O}_K,\ \mathfrak{p} \nmid \mathfrak{N},\ N(\mathfrak{p}) \le P\}$$

acting on the 26-dimensional block of V_{600}, with a common eigenform $\psi \in V_{\min}$, eigenvalues $a_\mathfrak{p}$, satisfying simultaneously the admissibility classes (A1)–(A7) of Section 4.

Then the Dirichlet series $L(s, \psi)$ tiled from $\mathcal{F}_P$ by Theorem 2.2:

1. *agrees with $L_{\text{sub}}(s) = \zeta_K(s)\,\zeta_K(s-1)\,C_2(s)$ up to the finite primes above $\mathfrak{N}$;*
2. *is a genuine cuspidal Hilbert modular L-function on $\mathrm{GL}_2(\mathbb{Q}(\sqrt{5}))$.*

Remark 5.2 (Important scope correction)**.** An earlier version of this theorem asserted a third conclusion — that the tiled L-function *satisfies the Riemann Hypothesis*. **That conclusion is withdrawn.** As shown by the explicit construction of §8 and the companion file `route_b/admissibility.py`, the admissibility classes (A1)–(A7) certify that $L(\psi, s)$ is a genuine *automorphic* L-function, but they do *not* imply that its zeros lie on the critical line. RH for $L(\psi, s)$ is precisely GRH for the corresponding object (here, an elliptic curve over $\mathbb{Q}(\sqrt{5})$), which is open. The Weil explicit formula is a valid identity relating zeros to primes, but it is a tool equivalent to information about the zeros — it does not force them onto the line. The reduction below is therefore a reduction of the *L-function* to finite data, not a reduction of *RH* to finite data.

5.2 Proof outline

Proof. **(1) Identification with L_{sub}.** By (A1)–(A2)+(A4), $\mathcal{F}_P$ generates a Dirichlet series with the correct functional equation and Galois-equivariance, matching the symbolic L_{sub} on its prime factorisation away from $\mathfrak{N}$. Equality on prime factors plus completion by the local factors at $\mathfrak{N}$ gives identity up to those finitely many primes.

(2) Cuspidal automorphy. By (A6), the eigenvalues are strictly Ramanujan-bounded, so the L-function is not Eisenstein. The Hecke-eigenform ψ on $V_{\min}$ with the multiplicativity of (A1) defines an automorphic representation of $\mathrm{GL}_2(K)$ by the converse theorem (Weil-Jacquet-Langlands), since the L-function satisfies the functional equation (A2) and grows polynomially.

(3) The explicit formula — and why it does *not* give RH. By Lemma 3.1, the Weil explicit formula computed from $\mathcal{F}_P$ converges to the archimedean side as $P \to \infty$, giving, for every admissible test h,

$$\sum_n h(\gamma_n) = (\text{archimedean}) - (\text{prime side via tiling}).$$

By (A6) the prime side is real for real h, so the zero sum is real — but *this is automatic for any self-dual automorphic L-function and says nothing about the location of individual zeros.* The explicit formula is an identity equivalent to information about the zeros; it does not constrain them to the critical line. Sato-Tate (A7) governs the distribution of the $a_{\mathfrak{p}}$ and likewise does not pin the zeros. Hence (A1)–(A7) establish that $L(\psi, s)$ is a genuine cuspidal automorphic L-function (conclusions 1–2), and **RH for it remains exactly as open as GRH**. See Remark 5.2. □

5.3 Conditional status

What is and is not proved

Theorem 5.1 is **conditional** on the existence of a finite-prime substrate Hecke system satisfying (A1)–(A7). The substrate framework provides:

- **Proved unconditionally:** the identity $L_{\text{sub}} = \zeta_K \cdot \zeta_K(\cdot - 1) \cdot C_2$; the existence and uniqueness of $W = V_{\min}$; the closure-flow attractor structure; the engine's verifier infrastructure.

- **Validated computationally at proxy level $N = 31$:** (A1) SM, (A3) DN, (A6) CB after Eisenstein projection, (A7) ST with finite-sample χ^2 tolerance.

- **Required to discharge:** a finite-prime Hecke system on the *genuine icosian setting at* $\mathfrak{N} \subset \mathfrak{O}_K$, not the proxy. This requires either (a) constructing cuspidal Hilbert modular forms over $\mathbb{Q}(\sqrt{5})$ at the substrate's level $\mathfrak{N}$ by classical means and verifying (A1)–(A7), or (b) a structural identification of the substrate's closure-flow eigenvalues with HMF eigenvalues by McKay-style correspondence.

Neither route has been completed in this bundle. The theorem isolates the finite remaining task.

6 The finite verification path

6.1 What needs to be computed

Let $\mathfrak{N} = \mathfrak{p}_{31} \subset \mathfrak{O}_K$ be the norm-31 prime level for $K = \mathbb{Q}(\sqrt{5})$. For each prime ideal $\mathfrak{p} \subset \mathfrak{O}_K$ coprime to $\mathfrak{N}$ with $N(\mathfrak{p}) \le P$, the verification computes:

(V1) **Brandt matrix.** The Brandt matrix $B_{\mathfrak{p}}$ acting on the space of theta series of the quaternion algebra ramified at $\mathfrak{N} \cdot \infty_1\infty_2$ over K.

(V2) **Eisenstein projection.** Subtract the Eisenstein eigenvalue $N(\mathfrak{p}) + 1$ to obtain the cuspidal part.

(V3) **Witness restriction.** Restrict to the 26-dim block of the substrate's τ-fixed eigenspace.

(V4) **Identification with closure-flow.** Verify that the resulting eigenvalues match $\mathcal{C}_\varphi$'s spectral data on $V_{\min}$.

(V5) **Run engine v0.6 verdict.** Submit $\mathcal{F}_P$ to the engine; collect verdict on (A1)–(A7).

6.2 Why this is a finite task

The number of generators required for any desired explicit-formula precision ϵ is bounded by Lemma 3.1:

$$P_0(h, \epsilon) = O\left(\left(\log \frac{1}{\epsilon}\right)^{1+1/r}\right)$$

for $h \in C_c^r$. For $r = 4$ and $\epsilon = 10^{-6}$ this is $P_0 \leq 200$, giving at most $\pi_K(200) = 46$ generators in $\mathcal{F}_{200}$.

6.3 Computational obstacles

The two real obstacles to executing V1–V5:

1. **SAGE Hecke support over $\mathbb{Q}(\sqrt{5})$.** Current SAGE (10.9 in our environment) provides `BrandtModule(N,1)` only over $\mathbb{Q}$. Hilbert modular Hecke at level $\mathfrak{N} \subset \mathfrak{O}_K$ is not available; `maximal_order` on `QuaternionAlgebra(K,-1,-1)` raises NotImplementedError. The proxy uses level $N = 31$ over $\mathbb{Q}$, which approximates but does not equal the genuine icosian Brandt matrices.

2. **Identification proof.** Even granted the Brandt matrices in (V1), the identification (V4) between Brandt eigenvalues and closure-flow eigenvalues on $V_{\min}$ is a non-trivial mathematical claim. It is plausible by McKay correspondence (the binary icosahedral group $2I$ is the icosian's automorphism, and its representation theory parametrises the cuspidal block), but it has not been proved.

The first obstacle is purely computational and can be removed by re-implementing Hilbert modular Brandt matrices over $\mathfrak{O}_K$ (a multi-month engineering project). The second is a genuine theorem to prove.

6.4 What the engine currently demonstrates

At level $N = 31$ over $\mathbb{Q}$ (proxy):

- (A1) SM: **passes** (multiplicativity verified on 15 primes)
- (A2) FE: **passes** (functional equation checked symbolically)
- (A3) DN: **passes** (Dirichlet normalisation)
- (A4) GUE: **N/A** at $\mathbb{Q}$-level proxy (Galois trivial)
- (A5) EF: **partial** (truncated explicit formula converges within tolerance, full convergence requires higher P)
- (A6) CB: **passes after projection** (cuspidal bound holds on the Eisenstein-projected eigenvalues)
- (A7) ST: **passes** (Sato-Tate χ^2 within $p > 0.1$ tolerance at $n = 15$ samples)

Five-of-seven at proxy level is the strongest verdict the substrate translation engine has yielded. At genuine icosian level, all seven would be required.

Honesty caveat: the proxy "passes" are not independent confirmation

A provenance audit of the engine code (companion file `CIRCLE_TEST.md`) found that the proxy verdicts above are *not* evidence that the substrate supplies genuine Hecke data:

- In the verdict-layer construction, the prime→eigenvalue assignment is *hardcoded* from the arithmetic targets $2\sqrt{p}$ (Ramanujan) and $\sqrt{1+p^2}$ (Eichler–Brandt); the substrate eigenvalues are not consumed. So (A6) and (A7) pass *tautologically* — the numbers were defined to satisfy those bounds.

- In the hybrid construction, the spectrum is obtained by least-squares fitting a feature library to the target Riemann zeros, i.e. to the answer; and even so the spectral match to $\{\gamma_n\}$ is **broken** (RMSE $\gtrsim 56$). The "HILBERT–POLYA PARTIAL" label was carried by weak distributional checks, not by reproducing the zeros.

What *is* genuine and substrate-derived is the 26-dimensional A_1-Galois block of V_{600} and the operator C_φ. What is missing is a parameter-free *geometric* prime→Hecke map on that block. The honest non-circular test is to compute the icosian ring's own intrinsic Brandt (neighbour) action — which is simultaneously geometric and the genuine Hilbert-modular Hecke operator — and check, *out of sample and without fitting*, whether its eigenvalues on the block reproduce the independently-tabulated norm-31 Hilbert newform. Until that computation exists, the hypothesis of Theorem 5.1 is wide open, not nearly discharged.

7 Relation to the other tilings

7.1 $W(H_4)$ does not extend to L-function statements

The geometric tiling $W(H_4)$ on V_{600} relates substrate vertices to each other. It does not naturally act on the L-function's zero set, because the zero distribution along the critical line is not finite or $W(H_4)$-orbit structured. So the geometric tiling does not directly attack RH.

7.2 Cascade tiling extends substrate, not zeros

Each cascade level adds substrate particles, refining coverage of the strip topology. As shown in [5], this gives a gradient-coverage metric that grows from $\sim$65% at V_{600} to $\sim$91% at $E_8 \times E_8$ and 100% only in the infinite limit. Like $W(H_4)$, cascade tiling does not in itself force zeros onto the critical line.

7.3 Hecke tiling is the unique L-function tiling

The Hecke tiling is the only one of the three that:

- has a finite generating set ($\mathcal{F}_P$ for any P),
- determines all infinite spectral data ($\{a_\mathfrak{n}\}_\mathfrak{n}$),
- connects to the L-function's zero set via the explicit formula.

This is why the shortcut to RH runs through Hecke, not through geometry or cascade. The substrate's prior bundles establish the geometric and cascade structure; this bundle isolates the Hecke layer as the analytical attack surface.

8 Status, scope, and open items

8.1 Update: the construction was carried out (2026-05-30)

The conditional hypothesis of Theorem 5.1 has since been *realised* at level $\mathfrak{p}_{31}$, not just posited. Using the icosian ring as the maximal order of the definite quaternion algebra $(-1, -1/\mathbb{Q}(\sqrt{5}))$ (Demblé [10]), the companion code `route_b/` computes the substrate's own Brandt/Hecke action at level $(5\varphi - 2)$ via the class-number-one $\mathbb{P}^1(\mathbb{F}_{31})$ model and finds:

- the unit group $2I$ reduces to $A_5 \subset \mathrm{PGL}_2(\mathbb{F}_{31})$ with orbits of sizes 12 and 20, so $\dim = 2$ (1 Eisenstein +1 cusp);
- the cuspidal Hecke eigenvalues $a_\mathfrak{q}$ **exactly match** the genuine norm-31 Hilbert newform (computed independently in `sims/sim_genuine_eigenvalues.py`) on all 11 prime ideals with $N(\mathfrak{q}) \leq 41$ — including signs and split-prime pairings, out-of-sample, with no fitting;
- admissibility (A1) multiplicativity $a_{\mathfrak{q}^2} = a_\mathfrak{q}^2 - N(\mathfrak{q})$, (A6) Ramanujan, and (A7) Sato-Tate are verified on this genuine data.

Thus the substrate-side identification (O2) is **confirmed** (not merely plausible): the substrate genuinely computes these Hecke eigenvalues. However, as Remark 5.2 and `route_b/admissibility.py` make explicit, this yields a bona fide automorphic L-function but *not* RH for it. The honest net effect is a verified arithmetic identification plus a corrected, weaker conclusion.

8.2 What is unconditional

- Theorem 2.2 (multiplicativity tiling): proved unconditionally as a statement about Hecke-multiplicative systems satisfying Ramanujan.
- Lemma 3.1 (explicit formula bridge): proved unconditionally modulo the standard explicit-formula machinery for Hilbert modular L-functions.
- Corollary 3.2: follows from Theorem 2.2 and Lemma 3.1.
- The seven admissibility classes (A1)–(A7) are well-defined and the verifier infrastructure is implemented in [6].

8.3 What is conditional

- Theorem 5.1 (the main result): conditional on the existence of a substrate Hecke system on the 26-dim block of V_{600} satisfying (A1)–(A7) at the genuine icosian level, not the proxy.
- The identification of substrate closure-flow eigenvalues with Hilbert modular Hecke eigenvalues is plausible by McKay correspondence but not proved.

8.4 Open items

(O1) Construct cuspidal Hilbert modular forms over $\mathbb{Q}(\sqrt{5})$ at the substrate's level $\mathfrak{N} = \mathfrak{p}_{31}$ (norm 31) by direct quaternionic Brandt matrix computation. Requires implementing HMF-Brandt over $\mathfrak{O}_K$ (Voight [9] has the algorithms; no SAGE implementation exists).

(O2) Prove that the substrate's closure-flow eigenvalues on $V_{\min}$ equal Hecke eigenvalues of the resulting HMF. Likely route: McKay correspondence on the binary icosahedral group $2I \subset \mathrm{SU}(2)$ with the cuspidal block as the non-trivial McKay-graph projector.

(O3) Run the engine v0.6 verifier against the constructed $\mathcal{F}_P$ and confirm (A1)–(A7) pass.

(O4) **Corrected.** Even with (O1)–(O3) discharged (which they now are at level $\mathfrak{p}_{31}$, see §8), Theorem 5.1 yields a genuine automorphic L-function, *not* RH. RH for it is GRH and remains open (Remark 5.2). The originally-stated "RH follows" was an overclaim and is withdrawn.

8.5 What this paper does *not* claim

1. No unconditional proof of RH.
2. No claim that the substrate framework alone, without the HMF identification in (O2), forces RH.
3. No claim about classical RH for $\zeta(s)$ separately from the L_{sub} identification. (Since $L_{\text{sub}} = \zeta_K(s)\zeta_K(s-1)C_2(s)$ and $\zeta_K = \zeta \cdot L(s, \chi_5)$, RH for L_{sub} implies RH for ζ at certain shifts and for $L(s, \chi_5)$, but the implication back to classical RH requires the additional step of identifying which zeros of L_{sub} come from ζ.)

9 Conclusion

The substrate's three symmetry structures — $W(H_4)$, cascade, Hecke — play different roles. Only Hecke tiling reaches the L-function's analytic content, because only Hecke has a finite fundamental domain and multiplicative extension. The Fractal Tiling Theorem isolates the precise finite mathematical task whose discharge would force RH: construct a cuspidal Hilbert modular form over $\mathbb{Q}(\sqrt{5})$ at level $\mathfrak{N} = \mathfrak{p}_{31}$ (norm 31) whose Hecke eigenvalues on small primes agree with the substrate's closure-flow eigenvalues on the 26-dim τ-fixed block of V_{600}, and verify the engine's seven admissibility classes pass.

The substrate framework has reduced RH from *"open in $\mathbb{C}$"* to *"open in a 46-dim algorithmic verification at finite primes plus a McKay-correspondence identification."* This is the most refined formulation the V_{600} programme has produced.

References

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